Research Article

Identification of Cervical Radiculopathies
Optimizing the Electromyographic Screen

Authors:

Timothy R. Dillingham, MD, MS
Tamara D. Lauder, MD
Michael Andary, MD
COL Shashi Kumar, MD
Liliana E. Pezzin, PhD
MAJ Ronald T. Stephens, MD
LTC Steven Shannon, MD

Affiliations:

From the Departments of Physical Medicine and Rehabilitation (TRD, TDL) and Emergency Medicine (LEP), the Johns Hopkins University, Baltimore, Maryland; the Department of Physical Medicine and Rehabilitation (MA), Michigan State University, East Lansing, Michigan; Physical Medicine and Rehabilitation Service (SK), Madigan Army Medical Center, Tacoma, Washington; Physical Medicine and Rehabilitation Service (RTS), Womack Army Medical Center, Fort Bragg, North Carolina; Physical Medicine and Rehabilitation Service (SS), Walter Reed Army Medical Center, Washington, DC.

Reprints:

All correspondence and requests for reprints should be addressed to Timothy R. Dillingham, MD, 10352 Waverly Woods Drive, Ellicott City, MD 21042.

Disclosures:

Funded, in part, by the 1994 Physical Medicine and Rehabilitation Education and Research Foundation Young Investigator Award.
The views expressed in this article are those of the authors and should not be construed to reflect the positions or views of the United States Army or the United States Government.

0894-9115/01/8002-0084/0
American Journal of Physical Medicine & Rehabilitation
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ABSTRACT

Dillingham TR, Lauder TD, Andary M, Kumar S, Pezzin LE, Stephens RT, Shannon S: Identification of cervical radiculopathies: optimizing the electromyographic screen. *Am J Phys Med Rehabil* 2001; 80:84–91.

Objective: To determine the optimal electromyography screening examination of the upper limb that ensures detection of those cervical radiculopathies, which can be electrodiagnostically confirmed, yet minimizes the number of muscles studied.

Design: A prospective multicenter study was conducted from May 1996 to September 1997 at five institutions. Patients who were referred to participating electrodiagnostic laboratories with suspected cervical radiculopathy were recruited. A standard set of muscles were examined by needle electromyography. Patients with electrodiagnostically confirmed cervical radiculopathies, based on electromyography findings, were selected for analysis. Muscle screens were tested against this group to determine whether the screen identified the patients with radiculopathy.

Results: There were 101 patients with cervical radiculopathies representing all cervical root levels. When paraspinal muscles were one of the screening muscles, five muscle screens identified 90% to 98% of radiculopathies, six muscle screens identified 94% to 99%, and seven muscle screens identified 96% to 100%. When paraspinal muscles were not part of the screen, eight distal limb muscles recognized 92% to 95% of radiculopathies.

Conclusion: This study demonstrated that six muscle screens including paraspinal muscles yielded consistently high identification rates. Studying additional muscles led to marginal increases in identification.

Key Words: Electrodiagnosis, Electromyography, Radiculopathy, Nerve Conduction, Cervical Region

The electrodiagnostic evaluation provides important information for assessing patients with upper-limb symptoms of pain, sensory loss, or weakness.[1–12] The differential diagnosis of these problems includes cervical radiculopathy, entrapment neuropathy, polyneuropathy, musculoskeletal disorders, or central nervous system causes such as myelopathy and stroke. The electrodiagnostician is compelled to perform a study sufficient to confidently identify or exclude cervical radiculopathy. Studies involving a large number of muscles, however, are uncomfortable to the patient. For this reason, delineating an optimal electromyography (EMG) screening examination that allows the examiner to identify a cervical radiculopathy (when one can be electrodiagnostically confirmed) yet minimizes the number of muscles studied to prevent excessive patient discomfort is of great clinical interest.[13]

The concept of a screening EMG encompasses detecting the possibility of an electrodiagnostically confirmable radiculopathy. If one of the muscles in the screen is abnormal, the screen must be expanded to exclude other diagnoses, and to fully delineate the radiculopathy level. Some radiculopathies cannot be confirmed by needle EMG.[6, 11, 12, 14] Radiculopathies that exclusively cause sensory root involvement, for example, will not produce abnormal EMG findings.[11, 12] If the rate of denervation is balanced by reinnervation in the muscle, then spontaneous activity is less likely to occur. Although EMG findings correlate with spinal imaging (myelography and magnetic resonance imaging) and intraoperative findings, there are some radiculopathies which demonstrate denervation potentials on EMG but show no anatomic structural reasons to explain this muscle denervation.[6, 15, 16] This has been termed chemical or inflammatory radiculitis.[15, 16] Conversely, patients can have a clear structural cause of nerve root compromise and yet have a normal EMG.[6, 14]

A screening EMG study involves determining whether the radiculopathy can be confirmed by EMG. If the radiculopathy cannot be confirmed, then presumably no amount of muscles can identify the radiculopathy. The process of identification can be conceptualized as a conditional probability. Given that a cervical radiculopathy can be confirmed by needle EMG, what is the minimum number of muscles which must be examined to confidently recognize or exclude this possibility?

Despite the great potential for minimizing the physicians' time and patients' discomfort, the optimal number of muscles for evaluating patients with limb symptoms has received little attention in the literature. Recently, we[13] retrospectively identified 175 electrodiagnostically confirmed cervical radiculopathies to assess this issue. Our analysis revealed that groups of seven muscles (including cervical paraspinal muscles) identified >94% of all radiculopathies. Although this analysis provided new insights into how many muscles must be studied, the retrospective nature of the study implied inherent limitations. Further, the sample comprised patients referred to a single institution, limiting the generalizability to a broader referral base. Finally and most importantly, there was no standard group of muscles studied for each patient. The lack of standardization implied that the investigators could not determine whether a screen failed to identify a patient with radiculopathy because the muscles in that screen were not frequently abnormal, or whether a particular muscle was simply not studied for all patients.

The purpose of this study was to determine the optimal EMG screening examination of the upper limb that ensures identification of those cervical radiculopathies which can be electrodiagnostically confirmed. By using a prospective multicenter design, this study overcomes the limitations of our previous analysis.

METHODS

Patient Selection. A prospective study was conducted between May 1996 and September 1997 at the following five medical centers treating diverse populations: the Johns Hopkins University, Baltimore, MD; Michigan State University, East Lansing, MI; Madigan Army Medical Center, Tacoma, WA; Womack Army Medical Center, Fort Bragg, NC; and Walter Reed Army Medical Center, Washington, DC. Patients who were referred for a suspected cervical radiculopathy to the electrodiagnostic laboratories in the Department of Physical Medicine and Rehabilitation at one of these medical centers were asked to participate. Subjects participated on a voluntary basis after signing written informed consent. The study protocol was approved by the appropriate Institutional Review Committee at each of the participating institutions.

Patient Evaluation. Standardized procedures and data collection sheets were implemented at all participating institutions. Patients completed a questionnaire, which collected detailed information on symptoms and past medical history. A standardized physical examination, performed on all subjects, consisted of the following: (1) a neurologic examination, including manual muscle testing, sensation (vibration and pinprick), and reflex assessments; (2) a musculoskeletal examination; and (3) special tests, such as Tinels and Phalens. After the questionnaire and physical examination, electrodiagnostic evaluations conducted by attending physicians or supervised resident physicians were performed for all patients. All co-investigators were certified by the American Board of Electrodiagnostic Medicine and were skilled musculoskeletal physicians; they ensured that the physical examination and electro-

diagnostic findings were interpreted uniformly across institutions.

The standardized electrodiagnostic study consisted of at least the following: (1) one upper-limb motor nerve conduction study, (2) one upper-limb sensory nerve conduction study, and (3) needle EMG with either monopolar or concentric needles. The American Association of Electrodiagnostic Medicine guidelines[17] for the evaluation of radiculopathy specify that a motor and sensory conduction should be performed to exclude the possibility of polyneuropathy. The motor and sensory nerves were not precisely specified. Ten proximal and distal muscles were part of the basic screening procedure and studied for all subjects (whenever clinically possible). They were chosen to represent all cervical radiculopathy levels based on accepted anatomic innervations.[18] Muscles which comprised the standard screen and the cervical root levels they represented included the following: cervical paraspinal muscles (PSM), posterior rami C5–T1), deltoid (C5–6), biceps (C5–6), triceps (C7–8), pronator teres (C6–7), extensor digitorum communis (C7–8), flexor carpi radialis (C6–7), flexor carpi ulnaris (C7–8), abductor pollicis brevis (C8–T1), and first dorsal interosseous (C8–T1). All data entry and coding, including scoring of abnormalities, were reviewed for accuracy and consistency. Information about additional nerves or muscles studied at the electromyographer's discretion to ensure an accurate diagnosis was also recorded.

Sample Selection. The sample used in these analyses consisted of all patients with an electrodiagnostically confirmed cervical radiculopathy. Cervical radiculopathies were defined by the presence of abnormalities in two or more muscles innervated by the same nerve root level but different peripheral nerves. Nonparaspinal upper-extremity muscles were considered abnormal if they demonstrated the following: (1) positive sharp waves; (2) fibrillation potentials; (3) complex repetitive discharges; (4) high-amplitude, long-duration motor unit action potentials; (5) increased polyphasic motor unit action potentials (>30%); or (6) reduced recruitment. Quantitative motor unit analysis was not performed; rather judgments about the size and morphology of motor units were qualitatively made by the investigators in this study. PSM were considered abnormal if they showed fibrillation potentials, positive sharp waves, or complex repetitive discharges. Patients were also included in the study if only the PSM showed abnormalities. The PSM were not considered for determining inclusion if the patient had a history of neck surgery. Increased insertional activity was not used for the inclusion criteria, but was recorded for each muscle studied. For patients with bilateral cervical radiculopathies, the side with the greatest number of muscles studied was chosen for analysis. Finally, subjects with demonstrated evidence of peripheral polyneuropathies, myopathies, or other generalized conditions were excluded from the study. Patients were included nonetheless if there were a radiculopathy and a focal entrapment neuropathy, such as carpal tunnel syndrome.

Individual Muscle and Muscle Screen Identifications. Individual muscle identification rates were determined by assessing the percentage of patients with cervical radiculopathy for whom the tested muscle was abnormal. For instance, if a particular muscle was abnormal for a patient with a cervical radiculopathy, then that patient was considered successfully identified. Muscles were then combined into screens. These screens contained muscles with innervation levels representing C5 to T1 nerve roots. The ability of the screens to identify a cervical radiculopathy was then tested. If any muscle in the screen was abnormal for a particular patient with cervical radiculopathy, then the screen successfully identified that patient.

Two types of muscle abnormalities were tested to assess their identification rates for both individual muscles and for muscle screens. A more restrictive category of muscle abnormality consisting of *spontaneous activity* was analyzed. For the purposes of this study, spontaneous activity was considered to include only fibrillations and positive sharp waves. A more inclusive *neuropathic* category reflecting other findings was also assessed. This neuropathic category included other EMG signs of muscle denervation, spontaneous activity, complex repetitive discharges, reduced recruitment, increased insertional activity, increased polyphasic potentials, or large-amplitude long-duration motor unit action potentials. A broader neuropathic category encompassing all neuropathic findings, although less specific, ensures that detection is optimized.

RESULTS

Subjects. There were 101 patients with electrodiagnostically confirmed cervical radiculopathies. Males comprised 59% of the sample. The average age was 50 yr (SD = 16 yr), range 16 to 88 yr. All subjects had abnormal muscles consistent with the inclusion criteria. At least one muscle had spontaneous activity for 88% of all patients with cervical radiculopathies. Two or more muscles showed spontaneous activity for 65% of the sample. This indicates that the majority fulfilled the inclusion criteria based on having fibrillations or positive sharp waves and not other more subjective neuropathic changes. Further, additional muscles besides the standard screen were examined by needle EMG in 64% of subjects, with the most commonly studied being the supraspinatus (31%), infraspinatus (29%), and the abductor digiti minimi (15%). This indicates that electromyographers used their clinical judgment and studied additional muscles beyond

the standard screen as dictated by the clinical circumstances.

Radiculopathy levels were defined by the group of muscles that were abnormal according to generally accepted anatomic innervation levels.[18] The minimum number of levels that explained all abnormal muscles defined the radiculopathy levels. In this sample of patients with electrodiagnostically confirmed cervical radiculopathies, 2% involved only the C5 level, 14% the C5–6 levels, 11% the C6 level, 13% the C6–7, 23% the C7 root level, 13% the C7–8 levels, 7% the C8 level, 10% the C8–T1 levels, 5% involved more than two levels, and 2% were of indeterminate levels (only the PSM were involved). Neck surgery had been performed before electrodiagnostic study for 7% of patients.

Sixty-five percent of patients reported arm pain and 62% reported neck pain. Based on a pain drawing, 55% of the sample also described suffering from radicular pain (pain extending from the neck to at least the forearm). Patients were also asked to describe their symptoms when they were given a standard set of symptom descriptor terms from which to choose. Eighty percent reported numbness, 73% tingling, 66% weakness, and 35% described burning sensations. These self-reported symptoms indicated that this group was composed of patients with a variety of complaints both sensory and motor, typical of an upper-limb symptom referral population seen in an electrodiagnostic laboratory.

TABLE 1
Individual muscle identification rates for 101 patients with electrodiagnostically confirmed cervical radiculopathies

Muscle	% Studied	Neuropathic (%)	Spontaneous Activity (%)
Standard Screen			
Paraspinals	96	56	47
Pronator teres	100	46	31
Triceps	100	38	21
Flexor carpi radialis	99	37	25
Extensor digitorum communis	99	36	22
Abductor pollicus brevis	97	30	16
Flexor carpi ulnaris	100	28	16
Biceps	100	24	11
Deltoid	100	23	15
First dorsal interosseus	98	23	11
Additional Muscles			
Infraspinatus	29	45	31
Supraspinatus	31	13	13

For the neuropathic findings column, the patient with radiculopathy was identified if the muscle showed positive waves; fibrillations; increased polyphasic potentials; neuropathic recruitment; increased insertional activity; complex repetitive discharges (CRDs); or large-amplitude, long-duration, motor-unit action potentials. For paraspinal muscles, the neuropathic category only included fibrillations, increased insertional activity, positive waves, and CRDs. In the spontaneous activity column, the muscle had to demonstrate fibrillations or positive sharp waves. Standard screen muscles were those prospectively studied for the majority of patients. Additional muscles were those studied at the discretion of the electromyographer in >20 patients with radiculopathies. Increased insertional activity was the sole neuropathic finding for <3% of the muscles.

Individual Muscle Identifications. Table 1 shows identification rates for each individual muscle. The PSM identified the majority (56%) of radiculopathies. For each muscle, when utilizing the neuropathic definition, the identification rates were approximately 10% to 15% higher than when only spontaneous activity defined an abnormal muscle. Increased insertional activity was the sole neuropathic finding for <3% of the muscles. Over 96% of subjects had all muscles studied, indicating that, for the majority of patients, these muscles were accessible.

Muscle Screen Identifications. The ability of different muscle screens to identify electrodiagnostically confirmed cervical radiculopathies is shown in Tables 2 through 5. These muscle screens consisted of various permutations of muscles in the standard screen such that all root levels would be represented. A number of different combinations were evaluated; the combinations presented in the tables represent the range of detection rates for the presence of an electrodiagnostically confirmable radiculopathy.

Table 2 shows the identification rates for five muscle screens. When the more inclusive neuropathic definition was used, identification rates were substantially higher than if only spontaneous activity was considered. Likewise, when muscle screens did not include PSM, identification rates were lower than when the screen contained PSM. For five muscle screens that included the paraspinals, identification rates ranged from 90% to 98% with neuropathic findings and 73% to 80% when only spontaneous activity was considered. There were no substantial differences in identification rates for five muscles screens in a subset of patients with radicular pain compared with those with nonradicular pain.

As shown in Table 3, six muscle screens yielded identifications of 94% to 99% when PSM were included. Without the PSM, 87% to 94% were identified. These findings indicated that six muscle screens provided excellent detection for

TABLE 2
Five muscle screen identifications of patients with cervical radiculopathies

Muscle Screen	Neuropathic (%)	Spontaneous Activity (%)
Without paraspinals		
Deltoid, APB, FCU Triceps, PT	92	65
Biceps, triceps EDC, FCR, FDI	85	54
Deltoid, triceps EDC, FDI, FCR	84	58
Biceps, triceps PT, APB, FCU	91	60
With paraspinals		
Deltoid, triceps, PT APB, PSM	98	80
Biceps, triceps, EDC FDI, PSM	95	73
Deltoid, EDC, FDI PSM, FCU	90	73
Biceps, FCR, APB PT, PSM	95	77

APB, abductor pollicis brevis; FCU, flexor carpi ulnaris; FCR, flexor carpi radialis; PSM, cervical paraspinal muscles; FDI, first dorsal interosseous; PT, pronator teres; supra-supraspinatus, infra-infraspinatus; EDC, extensor digitorum communis.

The screen detected the patient with cervical radiculopathy if any muscle in the screen was one of the muscles which were abnormal for that patient. Neuropathic findings for nonparaspinal muscles included positive waves; fibrillations; increased polyphasic potentials; neuropathic recruitment; increased insertional activity; complex repetitive discharges (CRDs); or large-amplitude, long-duration, motor-unit action potentials. For paraspinal muscles, the neuropathic category included fibrillations, increased insertional activity, positive waves, or CRDs. Spontaneous activity referred only to fibrillations or positive sharp waves.

an electrodiagnostically confirmable cervical radiculopathy.

Seven muscle screen identifications are shown in Table 4. These screens identified 96% to 100% of the sample. By adding one muscle over a six muscle screen, only slightly higher identifications were achieved. The 100% identifications by several of these seven muscle screens required the inclusion of the supraspinatus. This was a result of the presence of one patient in the sample who displayed abnormalities in only the rhomboid and supraspinatus muscles. The supraspinatus and rhomboid were additional muscles studied because of the electrodiagnostician's clinical suspicion.

Eight muscle screen identifications are shown in Table 5. Identification rates with PSM were very high (97% to 100%). If the PSM were not included, identifications ranged from 92% to 95%. Nine and ten muscle screens without PSM did not offer any improvements in identification rates over those of eight muscle screens without paraspinals. This suggests that whenever the PSM are not reliable (for example, in patients with previous cervical spine surgery), then eight upper-limb muscles should be studied.

DISCUSSION

Determining how many and which muscles to study to successfully detect a cervical radiculopathy (when one can be electrodiagnostically confirmed) is an important clinical decision. The results of this study indicate that if six muscles representing all cervical root levels and including the cervical PSM are studied, then the examiner can be confident of detecting a cervical radiculopathy, which can be confirmed by EMG. Adding additional muscles results in only marginal increases in identification, although 100% identification is achieved with several seven muscle screens. Studies involving eight muscles provide no better identification than seven muscles. If the PSM cannot be reliably studied,[7, 8, 19, 20] then eight limb muscles are necessary to identify >92% of cervical radiculopathies. This study begins to clarify for electrodiagnosticians the point of diminishing returns, beyond which diagnostic certainty for detecting a cervical radiculopathy is not enhanced by examining more muscles.

It is important to note that if an electrodiagnostician can only study five muscles (including paraspinals) because of limited patient tolerance, only a few electrodiagnostically confirmable cervical radiculopathies will be missed; identification based on five muscle screens (including paraspinals) ranged from 90% to 98%.

Tables 2–5 provide the electrodiagnostician with useful muscle screens. These screens yield high detection rates, utilize muscles that are readily localized, and minimize patient discomfort. Useful combinations of different muscles are important to know, because some muscles may be inaccessible for some patients as a result of wound dressings, casts,

TABLE 3
Six muscle screen identifications of the patients with cervical radiculopathies

Muscle Screen	Neuropathic (%)	Spontaneous Activity (%)
Without paraspinals		
Deltoid, APB, FCU Triceps, PT, FCR	93	66
Biceps, triceps, FCU EDC, FCR, FDI	87	55
Deltoid, triceps EDC, FDI, FCR, PT	89	64
Biceps, triceps, EDC PT, APB, FCU	94	64
With paraspinals		
Deltoid, triceps, PT APB, EDC, PSM	99	83
Biceps, triceps, EDC FDI, FCU, PSM	96	75
Deltoid, EDC, FDI PSM, FCU, triceps	94	77
Biceps, FCR, APB PT, PSM, triceps	98	79

For identification criteria and definitions, see Table 2.

TABLE 4
Seven muscle screen identifications of patients with cervical radiculopathies

Muscle Screen	Neuropathic (%)	Spontaneous Activity (%)
Without paraspinals		
Biceps, triceps, FCU EDC, FCR, FDI, deltoid	89	62
Deltoid, triceps, APB EDC, FDI, FCR, PT	94	71
Biceps, triceps, EDC PT, APB, FCU, FDI	95	67
With paraspinals		
Deltoid, triceps, PT APB, EDC, PSM, biceps	99	83
Biceps, triceps, EDC FDI, FCU, PSM, deltoid	96	79
Deltoid, EDC, FDI PSM, FCU, triceps, APB	96	83
Biceps, FCR, APB, FDI PT, PSM, triceps	98	81
Deltoid, triceps, PT APB, EDC, supra, PSM	100	84
Biceps, triceps, supra, FDI PT, EDC, PSM	100	79
Biceps, triceps, EDC PT, supra, APB, PSM	100	81

For identification criteria and definitions, see Table 2.

arteriovenous shunts for dialysis, or other reasons.

These muscle screens alert the examiner to the possibility of a radiculopathy. The examiner must then expand and tailor the study to exclude other disorders such as polyneuropathy, motor neuron disease, or entrapment neuropathy. Likewise, additional muscles are usually needed to identify the actual electrodiagnostic root level.[21] The muscle screen provides the electrodiagnostician with a limited number of muscles, which, if normal, can reliably exclude an electrodiagnostically confirmable radiculopathy. Studying additional muscles is unwarranted because it provides no higher probability of identification.

Because of the screening nature of this evaluation, assessing more subtle findings of motor axonal neuropraxia or axonal loss such as reduced recruitment or polyphasic potentials is indicated. Although reducing specificity and interrater reliability, assessing for these subtle EMG signs ensures high detection rates. If such findings are recognized, the study must be expanded to determine whether these findings reflect a radiculopathy (if found in combination with other myotomal abnormalities), an entrapment neuropathy (if limited to a single nerve distribution), a generalized condition (if found in other limbs), or whether they are of no clinical significance. Only an experienced electrodiagnostician can render this judgment after performing a pertinent history and physical examination, and then completing a thorough electrodiagnostic study.

This study highlights the fact that screens involving more than seven muscles (including paraspinals) are not indicated unless clinical suspicion for cervical radiculopathy, brachial neuritis, suprascapular neuropathy, or other conditions is very high. These muscle screens nonetheless should not substitute for a clinical evaluation and differential diagnosis formulation by the electrodiagnostic consultant. Rather,

TABLE 5
Eight muscle screen identifications of patients with cervical radiculopathies

Muscle Screen	Neuropathic (%)	Spontaneous Activity (%)
Without paraspinals		
Deltoid, triceps, PT, APB FCU, EDC, biceps, infra	94	69
Biceps, triceps, FCU, APB EDC, FCR, FDI, deltoid	92	68
Deltoid, triceps, APB, biceps EDC, FDI, FCR, PT	94	71
Biceps, triceps, EDC, deltoid PT, APB, FCU, FDI	95	72
With paraspinals		
Deltoid, triceps, PT APB, EDC, PSM, biceps, FCU	99	85
Biceps, triceps, EDC, PT FDI, FCU, PSM, deltoid	99	83
Deltoid, EDC, FDI, biceps PSM, FCU, triceps, APB	97	84
Biceps, FCR, APB, FDI PT, PSM, triceps, deltoid	99	84
Biceps, triceps, supra, infra PT, EDC, PSM, APB	100	81

For identification criteria and definitions, see Table 2.

information from these screens allows the electrodiagnostician to tailor the EMG evaluation and make more informed clinical decisions regarding the probability of missing a radiculopathy when a given set of muscles are studied. Performing a focused history and physical examination is essential; these screens should not supplant such clinical assessment. The investigators in the study examined additional muscles 64% of the time, indicating that clinical assessments were made and that the patients were critically evaluated with an EMG study that was tailored to their individual problems. The investigators did not simply perform the standard muscle screen, and ignore the referral questions and patient presentations. For this reason, these findings are generalizable to a broad referral population. Screens with the standard muscles can identify 98% to 99% of patients supporting the usefulness of the muscles in the standard screen. The value of any additional muscle toward radiculopathy detection is minimal. However, as illustrated in Tables 4 and 5, 100% identification is achieved by including either the supraspinatus or infraspinatus to a group of standard muscles.

The heterogeneity of the population representing patients with a broad age range and with a variety of clinical symptoms underscores the applicability of these findings to the group of patients with upper-limb symptoms referred for electrodiagnostic examinations. These screening examinations detect the possibility of cervical radiculopathy for patients with neck or arm pain, and those with and without sensory and motor symptoms.

PSM have high identifications by themselves and substantially augment the detection rates for all screens.[13] For this reason, although they are technically more challenging to study, their inclusion is recommended in all screening examinations, whenever possible. If PSM cannot be studied, then eight limb muscles should be examined. Our inclusion criteria ensured that patients who had previous cervical spine surgery were not included in this study, based only on paraspinal findings, but rather demonstrated other abnormal non-PSM. Therefore, the results of this study are generalizable to patients with previous spine surgery, but require the examiner to study eight limb muscles to identify 92% of patients with radiculopathy.

Two important limitations of this study merit mentioning. First, standard muscles were chosen based on our previous study,[13] because they are easily identified, and because they are innervated by major upper-limb peripheral nerves and represent all cervical root levels. Other muscles may be equally as effective at identification. We chose to limit the examination to ten muscles based on our retrospective study.[13] It is reassuring that five and six muscle screens had such high identification rates. If they had not, we would be concerned that a ceiling effect had taken place; for instance, if nine or ten muscle screens were necessary to provide high identifications, it would be apparent that our standard screen should have included more than ten muscles. This, however, was not the case and six muscles identified most radiculopathies, further supporting the validity of our findings. Presumably by studying ten muscles in the standard screen, the investigators identified all cervical radiculopathies that could be detected with standard EMG. It is unlikely that studying more than ten muscles would have improved identification. Second, cervical spinal assessment with magnetic resonance imaging was not available for the majority of the sample. Magnetic resonance imaging is an important assessment tool which can provide valuable information about structural causes of radiculop-

athy. Correlation of imaging with electrodiagnostic findings utilizing six muscle screens may yield additional insights as to the identification of structural causes of radiculopathy, such as herniated nucleus pulposis and cervical spinal stenosis. This is an important area of future study.

The strength of this study follows from its multicenter, prospective nature and the utilization of a standard screen. This ensured the generalizability of our results to a typical referral population seen at most electrodiagnostic laboratories. The fact that six muscles correctly identified most radiculopathies in this large, heterogeneous sample is compelling evidence of the usefulness of these screens. Incorporating these findings into the overall clinical assessment and management of patients with neck and upper-limb symptoms and determining their usefulness in predicting treatment outcomes are important areas of future research.

CONCLUSIONS

Based on the findings from this study, six upper-limb muscles, including PSM, consistently identified most electrodiagnostically confirmable cervical radiculopathies. Studying additional muscles led to marginal increases in identification. Several seven muscle screens yielded 100% identification, and some five muscle screens identified >95%. When PSM are not reliable to study, as in the case of a post laminectomy patient, then eight distal muscles are necessary to achieve optimal identification.

ACKNOWLEDGMENTS

We thank the residents and fellows at the participating institutions for their assistance in data collection.

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